

CSC311 Introduction to Machine Learning Week 11 Exercises

When submitting problems, please **clearly indicate which question you are submitting**. Use the full question number shown below.

Training

W11-T1. Perform one iteration of the k-means clustering algorithm on the following data set

$$\mathbf{X} = \begin{bmatrix} 3 & 4 \\ 4 & 4 \\ 1 & 0 \\ 4 & 3 \\ 0 & 1 \\ 1 & 1 \end{bmatrix}$$

We will set $k = 2$, and initialize the cluster centers to $\begin{bmatrix} 2 & 2 \end{bmatrix}$ and $\begin{bmatrix} 3 & 3 \end{bmatrix}$.

W11-T2. In formalizing the k-means clustering algorithm, we showed that k-means clustering is an optimization process that minimizes a cost function. What is that cost function? Define each parameter in your answers.

W11-T3. Explain why the k-means clustering algorithm always converges.

W11-T4. Does the k-means clustering algorithm always converge to the global minima? Explain why or why not.

Validation

W11-V1. What implicit assumptions are made in the k-means clustering algorithm? How are these assumptions relaxed in a mixture model?

W11-V2. Suppose we have the following data set

$$x^{(1)} = 5, x^{(2)} = 10, x^{(3)} = 12, x^{(4)} = 4, x^{(5)} = 9$$

We wish to fit a mixture of two Gaussians to this data set, using Expectation-Maximization. In this model, the parameters $\mu_1, \mu_2, \sigma_1, \sigma_2$ are the means and standard deviations of the two Gaussians, and $\pi_1, \pi_2 = 1 - \pi_1$ are the cluster probabilities. Recall that the PDF of a univariate Gaussian is

$$p(x; \mu, \sigma^2) = \mathcal{N}(x; \mu, \sigma^2) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(x - \mu)^2}{2\sigma^2}\right)$$

(a) Suppose that we initialize

$$\mathbf{r}^{(1)} = \begin{bmatrix} 0.5 \\ 0.5 \end{bmatrix}, \mathbf{r}^{(2)} = \begin{bmatrix} 0.8 \\ 0.2 \end{bmatrix}, \mathbf{r}^{(3)} = \begin{bmatrix} 0.8 \\ 0.2 \end{bmatrix}, \mathbf{r}^{(4)} = \begin{bmatrix} 0.5 \\ 0.5 \end{bmatrix}, \mathbf{r}^{(5)} = \begin{bmatrix} 0.8 \\ 0.2 \end{bmatrix}$$

Given these responsibility vectors, estimate π_1 . You may use the MLE estimation formula without derivation.

- (b) Given the responsibility vectors from part (a), estimate μ_1 . Do not estimate the other parameters. Show all your work, but you may use the MLE estimation formula without derivation.
- (c) Suppose that we instead have the parameters $\mu_1 = 2, \mu_2 = 7, \sigma_1 = \sigma_2 = 1$, and $\pi_1 = \pi_2 = 0.5$. Given these parameter values, what is the value of the responsibility vector $\mathbf{r}^{(1)}$ for the data point $x^{(1)} = 5$? Show all your work.
- (d) Was the computation in each of parts (a), (b), and (c) part of the “Expectation” or the “Maximization” step of the E-M algorithm?

Generalization

W11-G1. The Poisson distribution is used to model data that consists of non-negative integers. Its probability mass function is $p(k; \lambda) = \frac{\lambda^k e^{-\lambda}}{k!}$. For the data set $k_1, k_2, \dots, k_n$, the maximum likelihood estimate for λ is $\frac{1}{n} \sum_{j=1}^n k_j$.

Suppose you observe the integers $\{x^{(1)}, x^{(2)}, \dots, x^{(N)}\}$ in your training set. Your model assumption is that each integer is sampled from one of two different Poisson distributions. You would like to learn this model using the EM algorithm.

- (a) List all the parameters of the model.
- (b) What is the likelihood of the training set under the model?
- (c) Derive the E-step for this model. Your answer should include a mathematical justification.